Exercise 1 – Introduction to Speech Recognition

Gal Noy (209346485)
Amit Ezer (316097575)

1 Question 1 – DFT and Windowing

(a)–(b) DFT of the First Signal

The signal contains $N = 400$ samples at $f_s = 16{,}000$ Hz, giving a duration of 25 ms and a frequency resolution of $\Delta f = f_s/N = 40$ Hz. We construct it as

$$s_1[n] = \sum_{i=1}^{3} A_i \cos\left(2\pi f_i \frac{n}{f_s} + \phi_i\right), \qquad 0 \le n < N.$$

The DFT is computed by multiplying the signal by the explicitly constructed matrix

$$W_{k,n} = e^{-j2\pi kn/N}, \qquad \mathbf{X}_1 = W\mathbf{s}_1.$$

Only bins $k = 0, \ldots, 199$ are displayed: for a real signal, $X[N-k] = \overline{X[k]}$, so the negative-frequency half is redundant (the Nyquist bin $k = 200$ is a separate boundary bin). All spectra use $20\log_{10}(|X[k]|/P)$, where P is the maximum coefficient magnitude of the original signal. Values below -80 dB are clipped for display; each window comparison uses the same P for both curves.

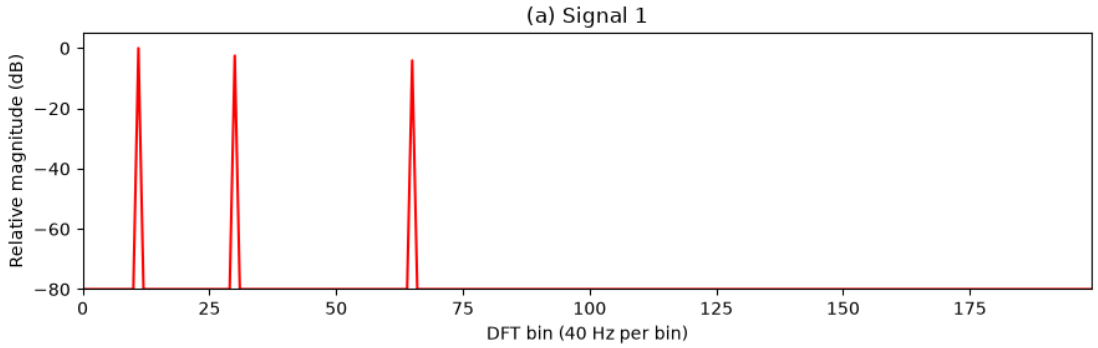


Figure 1: DFT magnitude of the first signal (relative dB).

(c) Amplitudes and Phases of the First Signal

A cosine splits its amplitude equally between the positive- and negative-frequency bins. Thus, for a bin-aligned tone, $|X[k]| = NA/2$, giving $A = 2|X[k]|/N$; its phase is $\phi = \arg X[k]$. All three frequencies are bin-aligned, yielding:

Tone (Hz)	Bin k	Amplitude	Phase (rad)
440	11	0.40000	0.78540
1200	30	0.30000	0.00000
2600	65	0.25000	-2.35619

(d)–(e) DFT, Amplitudes and Phases of the Second Signal

The new frequencies are 450, 1100, and 2550 Hz. Their DFT-bin positions are 11.25, 27.5, and 63.75, respectively. None is bin-aligned, so **spectral leakage** spreads the energy over neighboring bins instead of producing isolated peaks.

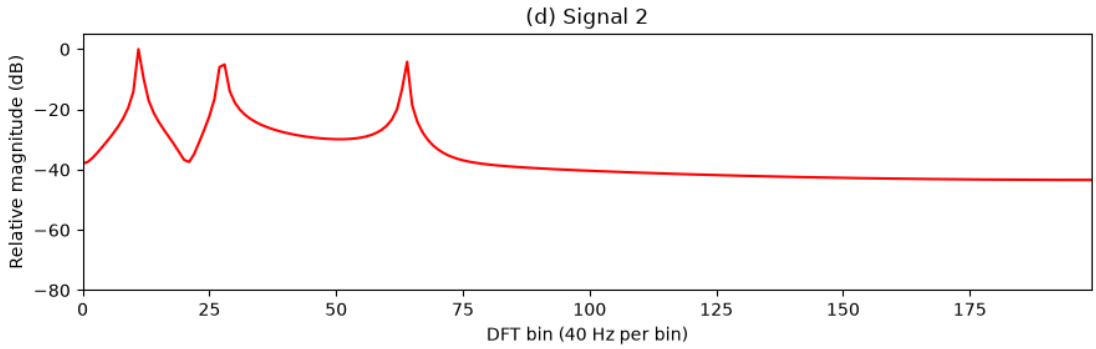


Figure 2: DFT magnitude of the second signal, showing spectral leakage (relative dB).

Using the closest bins (both equally close bins for 1100 Hz) gives:

Tone (Hz)	Bin k	Amplitude	Phase (rad)
450	11	0.35950	1.55805
1100	27	0.18144	1.54371
1100	28	0.20022	-1.54607
2550	64	0.22197	-3.11427

The selected bins correspond to 440, 1080, 1120, and 2560 Hz, respectively. These are amplitudes and phases of the DFT coefficients at those bin frequencies, not exact recovery of the original tone parameters.

(f) Hamming Window

Both signals are multiplied by the N-sample Hamming window

$$w[n] = \frac{25}{54} - \frac{21}{54}\cos\left(\frac{2\pi n}{N}\right), \qquad 0 \le n < N.$$

For the first signal, all tones are bin-aligned, so the original DFT has no spectral leakage. Windowing lowers the peaks and spreads energy into adjacent bins. For the second signal, which already has leakage, the window reduces the distant sidelobes but widens and lowers the main peaks. The Hamming window smoothly reduces the weight of samples near the frame edges, decreasing the discontinuity when the frame repeats and suppressing distant sidelobes for off-bin tones.

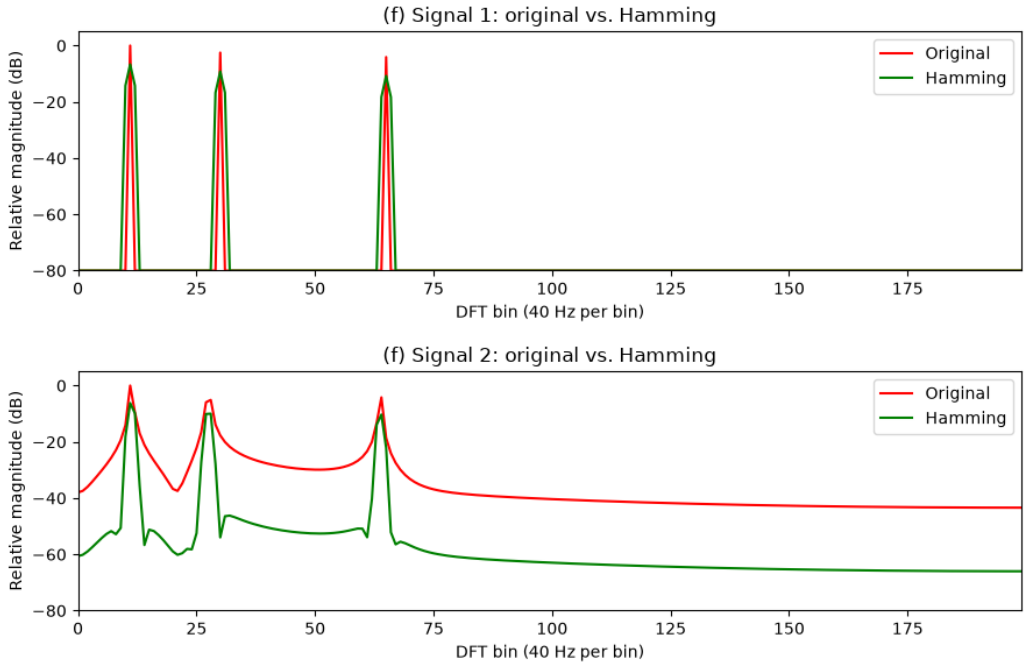


Figure 3: Original DFT (red) and windowed DFT (green) for each signal.

(g) Amplitudes and Phases after Windowing

The same coefficient-based estimates, now obtained from the windowed DFTs, are:

Signal	Tone (Hz)	Bin k	Amplitude	Phase (rad)
$s_1 w$	440	11	0.18519	0.78540
	1200	30	0.13889	0.00000
	2600	65	0.11574	-2.35619
$s_2 w$	450	11	0.17601	1.56887
	1100	27	0.11248	1.56756
	1100	28	0.11385	-1.56758
	2550	64	0.10980	-3.13751

For the first signal, the phases remain unchanged and the peak amplitudes are reduced by the average window weight, $\frac{1}{N}\sum_n w[n] = 25/54 \approx 0.463$. For the second signal, the window changes the distribution of energy between bins, affecting both the coefficient-based amplitudes and phases. The reported values refer to the same selected bin frequencies as in part (e).

2 Question 2 – Multivariate Gaussian Models

For M equally weighted vectors $\mathbf{x}_i \in \mathbb{R}^d$, we estimate

$$\boldsymbol{\mu} = \frac{1}{M}\sum_{i=1}^{M}\mathbf{x}_i, \qquad \Sigma = \frac{1}{M}\sum_{i=1}^{M}(\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^T.$$

The diagonal model retains only the diagonal of Σ. For both models, with normalization constant $C = (2\pi)^{-d/2}|\Sigma|^{-1/2}$,

$$\log p(\mathbf{x}) = \log C - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T\Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}).$$

The inverse covariance and $\log C$ are computed once during initialization.

(a) Models Estimated from `data_1.csv`

Both models have mean $\boldsymbol{\mu}_1 = (-0.098231,\ -0.097654,\ 0.084253)^T$. The diagonal covariance is represented by the variance vector

$$\boldsymbol{\sigma}_1^2 = (11.300731,\ 1.480702,\ 6.454034),$$

while the full covariance is

$$\Sigma_1 = \begin{pmatrix} 11.300731 & -0.032292 & 0.258590 \\ -0.032292 & 1.480702 & -0.079734 \\ 0.258590 & -0.079734 & 6.454034 \end{pmatrix}.$$

The normalization constants are $C_{1,\text{diag}} = 0.006109807$ and $C_{1,\text{full}} = 0.006114799$.

(b) Log-Likelihoods for `query_1.csv`

Query	Diagonal	Full
1	-9.598437	-9.595777
2	-6.439222	-6.397500
3	-7.255353	-7.226701
4	-6.604090	-6.562551
5	-5.701920	-5.692899
6	-5.308776	-5.309611
7	-7.417854	-7.458339
8	-7.273178	-7.289781
9	-5.601006	-5.583757
10	-7.302052	-7.275010

(c) Models Estimated from `data_2.csv`

Both models have mean $\boldsymbol{\mu}_2 = (5.298580,\ -0.881825,\ 1.290526)^T$. The diagonal covariance is represented by

$$\boldsymbol{\sigma}_2^2 = (8.886413,\ 6.460199,\ 3.886459),$$

and the full covariance is

$$\Sigma_2 = \begin{pmatrix} 8.886413 & 3.424745 & 1.619543 \\ 3.424745 & 6.460199 & -1.892735 \\ 1.619543 & -1.892735 & 3.886459 \end{pmatrix}.$$

The normalization constants are $C_{2,\text{diag}} = 0.004250764$ and $C_{2,\text{full}} = 0.006116646$.

(d) Log-Likelihoods for `query_2.csv`

Query	Diagonal	Full
1	-8.336550	-9.595454
2	-6.975472	-6.397200
3	-8.348692	-7.226405
4	-7.369369	-6.562250
5	-5.631708	-5.692596
6	-5.646387	-5.309307
7	-8.762895	-7.458044
8	-7.700626	-7.289480
9	-5.904931	-5.583454
10	-6.289201	-7.274711

(e) Explanation of the Differences

In the first dataset, the off-diagonal entries of Σ_1 are small relative to the variances, so the diagonal and full models give similar log-likelihoods. In the second dataset, the off-diagonal entries of Σ_2 are substantial. Ignoring these dependencies changes the diagonal-model scores considerably.

The two full models have nearly equal normalization constants, and the quadratic terms in the log-likelihood formula are also nearly equal for corresponding queries. Consequently, their log-likelihoods in (b) and (d) differ by at most 0.000324.